

Lab 1: SankofaPOWER – Foundations of GIS

Lab Instructions

In this lab we will download QGIS and make our first map!

Download & Install QGIS

This is best done on a computer as the mobile and tablet apps are primarily for field data collection.

You will need about 1 Gb of available storage on your computer.

1) Visit <https://qgis.org/download/>

a) Please find your operating system (OS) and **download** the Long Term Version (LTR) of QGIS.

b) Once downloaded, **install** the program. This may take some time.

c) If you run into issues, there is more documentation here:

<https://qgis.org/resources/installation-guide/>.

Your instructor will also be happy to assist!

Starting a QGIS Project

1) Organization is essential for GIS projects!

a) Open your [file manager](#) application and navigate to **Desktop**.

b) Create a new folder for this project called, **Sankofa GIS**

c) Within this new folder, create one more folder called, **Data**

2) Now, let's start QGIS. This can be done like any other application on your computer.

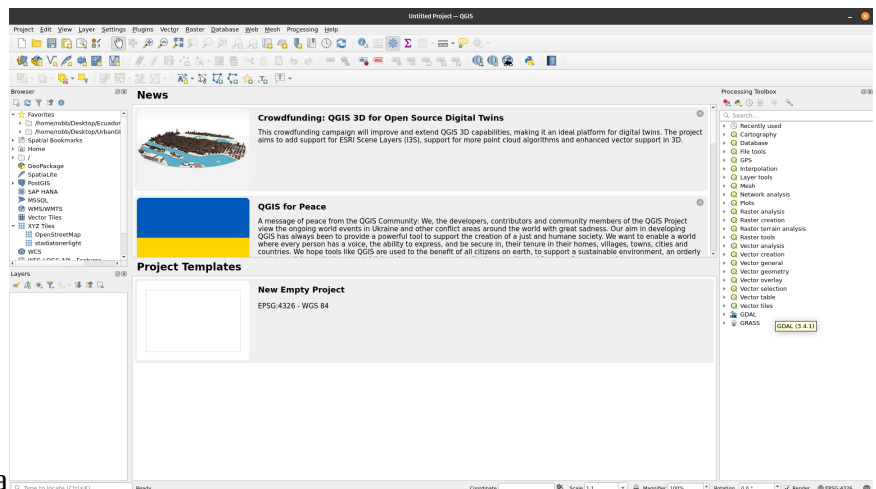
a) Your screen should look something like this ->

3) Finally, we will begin a new project

a) Double click on **New Empty Project** under Project Templates

b) Click on the **Project** tab at the top left and select **Save As**

c) Save this project in the Sankofa GIS folder as **Sankofa.qgz**



d) That's it for now, we'll be back once we have some data!

Creating a Point

In this section we will learn one way to create a [GeoJSON](https://geojson.io/).

1) Visit <https://geojson.io/> and click on the ? Help tab to learn more about this project

2) Head back to the `</> JSON` pane to watch as we build our data!

a) On the map, navigate to a place that is meaningful to you

b) Toggle the Draw Point tool. The cursor on the map should now look like crosshairs.

c) Click on the place. Did you see the point get added on the JSON pane?

3) Let's add some important [attributes](#) to our point.

a) Place your cursor in the squiggly brackets after "properties" in the `</>JSON` pane and click.

b) Edit the GeoJSON to include these important pieces of information:
name, date_of_visit, description.

The formatting is as follows:

```
"properties":  
{  
  "name": "My Juneteenth Interview",  
  "date_of_visit": "6/19/25",  
  "description": "This is where AMA conducted  
    interviews with folks to hear food stories."},
```

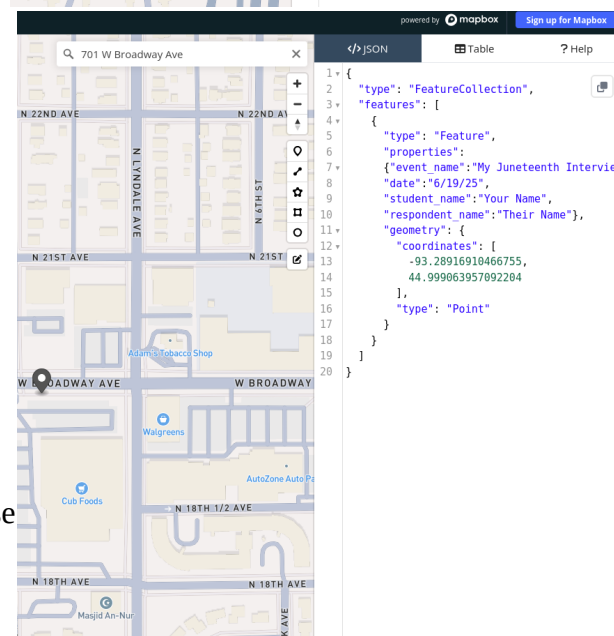
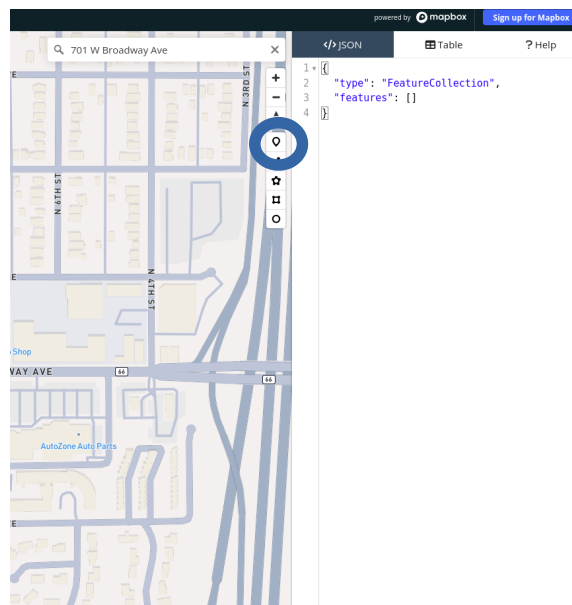
Please compare your screen to the image on the right to verify. The quotation marks (") are important!

4) Finally, let's download the data we've created.

a) On the top left of the map move your mouse to **Save | GeoJSON** and click. This should begin a download.

b) In your download folder, there should be a file called, map.geojson. **Rename to MyPlace.geojson**

c) Move **MyPlace.geojson** to the Sankofa GIS/**Data** folder



Let's Map!

In this section we will create a map to complement the place we documented.

Be sure to save often!! It can be done anytime by pressing **Ctrl S**

1) In QGIS, click on the **Layer** tab at the top of the screen. Select **Add Layer | Add Vector Layer**

- a) Under Source, click on the ..., navigate to MyPlace.geojson, and select open.
- b) Click the **Add** button.
- c) MyPlace should appear in the Layers Pane on the left of the screen. **Close** the Data Source Manager window.

2) We should see a dot sitting on a blank screen. Let's add a basemap!

- a) In the **Browser** pane on the left, scroll down to **XYZ Tiles** and click on the down arrow.
- b) Double Click **OpenStreetMap**. It should appear on the top of the Layers pane.
 - i) If it does not appear, right click on the OpenStreetMap layer and select **Zoom to Layer(s)** to allow the map to render in a lower resolution.
- c) On the layer pane, **Click and drag** the OpenStreetMap layer below the MyPlace Layer so it doesn't cover up our data.
- d) Lastly, right click on MyPlace and select **Zoom to Layer(s)**. Is it in the correct spot?

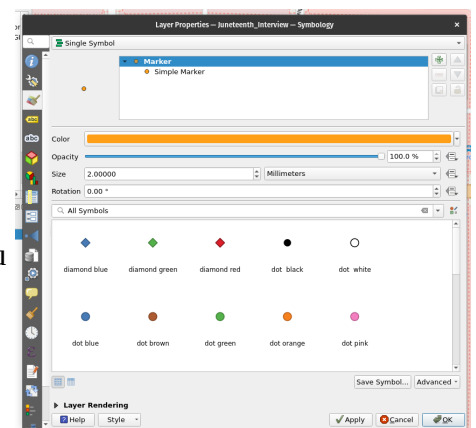
3) The basemap is a little blurry. This is because it is distorted by the projection of our project!

- a) Go to the **Project** tab and select **Properties**
- b) Click the **CRS** tab and type in the code **3857** then select **WGS 84 / Pseudo-Mercator** (EPSG:3857)
- c) Click **Apply** and **OK**.

4) Now for the fun part – Symbology!

- a) Right click on the MyPlace layer and select **Properties**
- b) Click the **Symbology** tab (See image)
- c) Change any of these options to make the point stand out! You can click **Apply** to see changes on the map. Once you're satisfied, click **OK**
- d) Feel free to experiment with **Labels** tab as well!

- i) My suggestion, set the **Value** of the label as **name**



5) Now that our layers are looking good, let's put the finishing touches together in a **Layout**

a) Select the **Print Layout** button

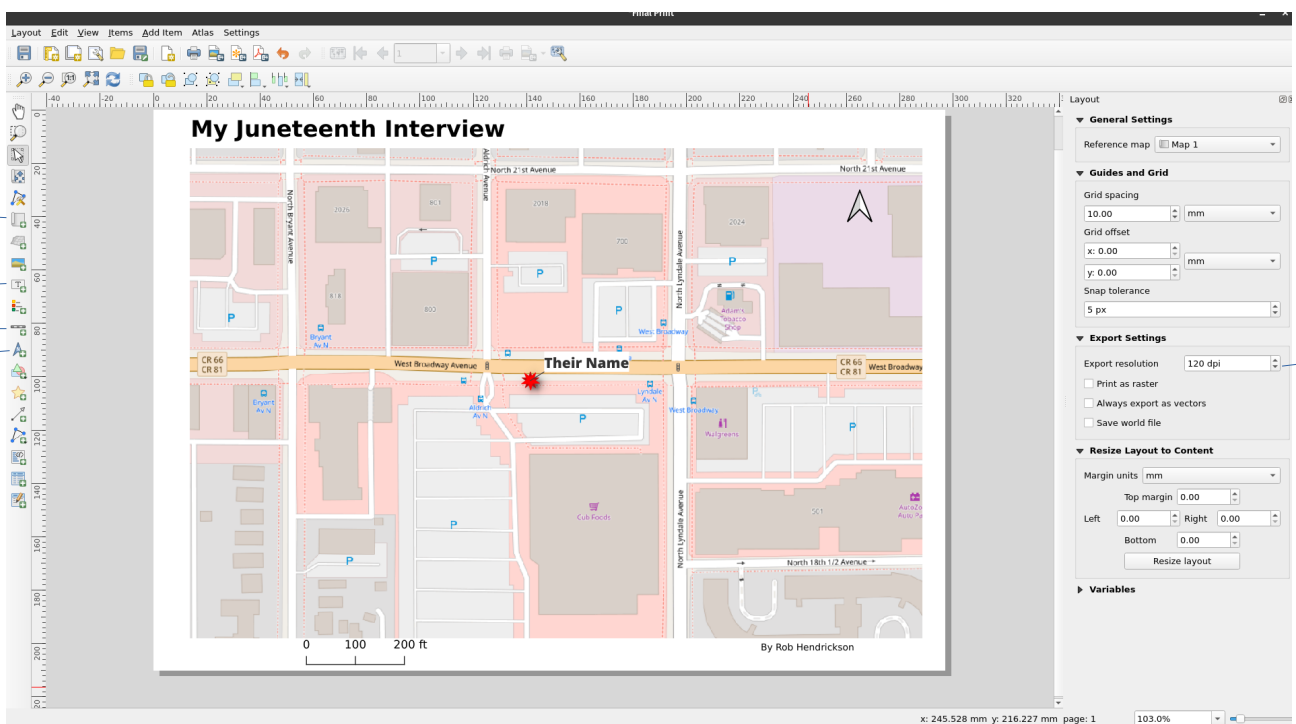
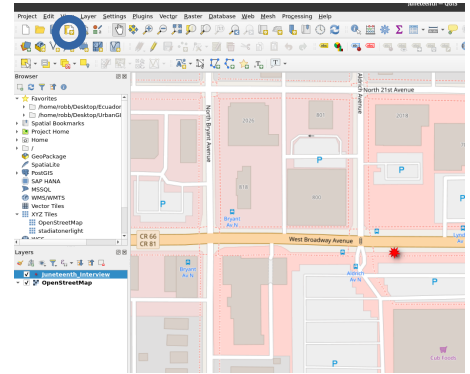


b) Title the layout **My First Layout** and select **OK**

For the following directions, please refer to the letters on the side of the image below.

c) On the left of your screen, select the **Add Map** tool. **Click and drag** a rectangle that covers most of the page and release.

Leave some space for a title, author, & scale bar!



d) Use the **Text Box** tool to create a title and author name

e) Use the **Scale Bar** tool to draw a scale bar at the bottom left of the map

f) Use the **North Arrow** tool to draw a north arrow at the top right of the map

g) For any of the objects listed above, you may right click and change the **Item Properties** to further customize.

h) In **Layout Properties**, set the **Export Resolution** to 120 dpi so the basemap renders correctly

i) When everything looks good, go to the **Layout** tab and select **Export as PDF**. Save it in the Sankofa GIS folder. Click through the final windows and select **Save**.

Save this PDF to your Google Drive and/or email it to instructors to ensure it doesn't get lost.

Don't forget to save your QGIS project!

